J1: 16-CH DIFFERENTIAL IF / BASEBAND INPUT BUS (Samtec QSH-030-01-L-D-A)

Pin 1/2: V_IN0_P/N

Pin 3/4: V_IN1_P/N

Pin 5/6: V_IN2_P/N

Pin 7/8: V_IN3_P/N

Pin 9/10: V_IN4_P/N

Pin 11/12: V_IN5_P/N

Pin 13/14: V_IN6_P/N

Pin 15/16: V_IN7_P/N

Pin 17/18: V_IN8_P/N

Pin 19/20: V_IN9_P/N

Pin 21/22: V_IN10_P/N

Pin 23/24: V_IN11_P/N

Pin 25/26: V_IN12_P/N

Pin 27/28: V_IN13_P/N

Pin 29/30: V_IN14_P/N

Pin 31/32: V_IN15_P/N

32-WIRE BALANCED ANALOG BUS
$R(t) = (a, b) \in \mathbb{S} \cong \mathbb{R}^{16}$ (100 Ω Diff)

32

SLICE_0: Differential Crossbar & Summing Unit ($k = 0$, Basis e_0 (Real Scalar)) — [See Sheet 2]
Mux A (U1_0: ADG1606) $\rightarrow i_A = 0 \oplus c$ | Mux B (U2_0: ADG1606) $\rightarrow i_B = 0 \oplus d$ | DPDT (U3_0, U4_0: ADG1611)
Summer (U6_0: AD8130 + RN1_0: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},0}(t) = \frac{1}{\sqrt{2}}\big(\sigma(0 \oplus c, c)V^{\pm}_{0\oplus c} + s_d\sigma(0 \oplus d, d)V^{\pm}_{0\oplus d}\big)$

32

SLICE_1: Differential Crossbar & Summing Unit ($k = 1$, Basis e_1 (Lower Im($\mathbb{O}$))) — [See Sheet 2]
Mux A (U1_1: ADG1606) $\rightarrow i_A = 1 \oplus c$ | Mux B (U2_1: ADG1606) $\rightarrow i_B = 1 \oplus d$ | DPDT (U3_1, U4_1: ADG1611)
Summer (U6_1: AD8130 + RN1_1: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},1}(t) = \frac{1}{\sqrt{2}}\big(\sigma(1 \oplus c, c)V^{\pm}_{1\oplus c} + s_d\sigma(1 \oplus d, d)V^{\pm}_{1\oplus d}\big)$

32

SLICE_2: Differential Crossbar & Summing Unit ($k = 2$, Basis e_2 (Lower Im($\mathbb{O}$))) — [See Sheet 2]
Mux A (U1_2: ADG1606) $\rightarrow i_A = 2 \oplus c$ | Mux B (U2_2: ADG1606) $\rightarrow i_B = 2 \oplus d$ | DPDT (U3_2, U4_2: ADG1611)
Summer (U6_2: AD8130 + RN1_2: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},2}(t) = \frac{1}{\sqrt{2}}\big(\sigma(2 \oplus c, c)V^{\pm}_{2\oplus c} + s_d\sigma(2 \oplus d, d)V^{\pm}_{2\oplus d}\big)$

⋮ Identical Hierarchical Instances SLICE_3 .. SLICE_6 ($k \in \{3, 4, 5, 6\}$) ⋮

32

SLICE_7: Differential Crossbar & Summing Unit ($k = 7$, Basis e_7 (Lower Im($\mathbb{O}$))) — [See Sheet 2]
Mux A (U1_7: ADG1606) $\rightarrow i_A = 7 \oplus c$ | Mux B (U2_7: ADG1606) $\rightarrow i_B = 7 \oplus d$ | DPDT (U3_7, U4_7: ADG1611)
Summer (U6_7: AD8130 + RN1_7: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},7}(t) = \frac{1}{\sqrt{2}}\big(\sigma(7 \oplus c, c)V^{\pm}_{7\oplus c} + s_d\sigma(7 \oplus d, d)V^{\pm}_{7\oplus d}\big)$

32

SLICE_8: Differential Crossbar & Summing Unit ($k = 8$, Basis e_8 (Doubling Gen)) — [See Sheet 2]
Mux A (U1_8: ADG1606) $\rightarrow i_A = 8 \oplus c$ | Mux B (U2_8: ADG1606) $\rightarrow i_B = 8 \oplus d$ | DPDT (U3_8, U4_8: ADG1611)
Summer (U6_8: AD8130 + RN1_8: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},8}(t) = \frac{1}{\sqrt{2}}\big(\sigma(8 \oplus c, c)V^{\pm}_{8\oplus c} + s_d\sigma(8 \oplus d, d)V^{\pm}_{8\oplus d}\big)$

32

SLICE_9: Differential Crossbar & Summing Unit ($k = 9$, Basis e_9 (Upper Im($\mathbb{O}$)e_8)) — [See Sheet 2]
Mux A (U1_9: ADG1606) $\rightarrow i_A = 9 \oplus c$ | Mux B (U2_9: ADG1606) $\rightarrow i_B = 9 \oplus d$ | DPDT (U3_9, U4_9: ADG1611)
Summer (U6_9: AD8130 + RN1_9: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},9}(t) = \frac{1}{\sqrt{2}}\big(\sigma(9 \oplus c, c)V^{\pm}_{9\oplus c} + s_d\sigma(9 \oplus d, d)V^{\pm}_{9\oplus d}\big)$

32

SLICE_10: Differential Crossbar & Summing Unit ($k = 10$, Basis e_{10} (Upper Im($\mathbb{O}$)e_8)) — [See Sheet 2]
Mux A (U1_10: ADG1606) $\rightarrow i_A = 10 \oplus c$ | Mux B (U2_10: ADG1606) $\rightarrow i_B = 10 \oplus d$ | DPDT (U3_10, U4_10: ADG1611)
Summer (U6_10: AD8130 + RN1_10: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},10}(t) = \frac{1}{\sqrt{2}}\big(\sigma(10 \oplus c, c)V^{\pm}_{10\oplus c} + s_d\sigma(10 \oplus d, d)V^{\pm}_{10\oplus d}\big)$

⋮ Identical Hierarchical Instances SLICE_11 .. SLICE_14 ($k \in \{11, 12, 13, 14\}$) ⋮

32

SLICE_15: Differential Crossbar & Summing Unit ($k = 15$, Basis e_{15} (Upper Im($\mathbb{O}$)e_8)) — [See Sheet 2]
Mux A (U1_15: ADG1606) $\rightarrow i_A = 15 \oplus c$ | Mux B (U2_15: ADG1606) $\rightarrow i_B = 15 \oplus d$ | DPDT (U3_15, U4_15: ADG1611)
Summer (U6_15: AD8130 + RN1_15: ERA-3AEB1001V ±0.05%) $\implies V^{\pm}_{\mathrm{ADC},15}(t) = \frac{1}{\sqrt{2}}\big(\sigma(15 \oplus c, c)V^{\pm}_{15\oplus c} + s_d\sigma(15 \oplus d, d)V^{\pm}_{15\oplus d}\big)$

160-BIT DIGITAL CROSSBAR CONTROL BUS (CTRL[159:0])

10b → V_ADC0_P/N

10b → V_ADC1_P/N

10b → V_ADC2_P/N

10b → V_ADC7_P/N

10b → V_ADC8_P/N

160-Bit GPIO

10b → V_ADC9_P/N

10b → V_ADC10_P/N

10b → V_ADC15_P/N

J2 & U_ADC1..2: 16-CH PHASE-COHERENT ADC BANK (2× TI ADS52J90 14-Bit, 100 MSPS ADCs)

Pre-ADC Jammer Cancellation Guarantee:
• Incoming +80 dB JSR Jammer ($V_{\mathrm{jam}} = \pm 5.0$ V_{pp} on J1) is algebraically annihilated in SLICE_0..15 prior to J2:
$\|V_{\mathrm{ADC}}(U_J \cdot W_J)\|_\infty < 1.9$ $\mathrm{mV_{pp}}$ (−68.38 dB Mean Null).
• ADS52J90 Full-Scale Range: 2.0 V_{pp} differential (±1.0 V).
• ADC Saturation / Clipping Rate: 0.00% (vs. 99.6% without Crossbar).
• Authentic Anti-Partner Satellite Signal U_{sat} receives +3.01 dB coherent array gain ($\|U_{\mathrm{sat}} \cdot W_J\|_2 = \sqrt{2}$).

U_ADC1 (ADS52J90 Ch 0..7) & U_ADC2 (ADS52J90 Ch 8..15)

• Differential Analog Inputs: INP1..INP8, INM1..INM8 (100 Ω term)
• Sample Clock: CLKP/CLKM = 65.536 MHz (Ultra-Low Jitter < 100 fs)
• Digital Output: 16× Serial LVDS / JESD204B Q1.14 Fixed-Point Lanes
• SNR: 71.5 dBFS | SFDR: 86 dBc | Power: 48 mW/ch

16× Q1.14 LVDS

U_CTRL: 256-BIT CAYLEY–DICKSON SIGN ROM & SINGLE-SNAPSHOT ($K = 1$) ASSOCIATOR CONTROLLER (Xilinx Artix-7 XC7A35T-1CPG236C / sedenion_zd_beamformer.v)

1. Single-Snapshot Associator Spoofer Detector ($K = 1$):
Computes $\mathcal{A}_{\mathrm{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (-2(a \times_{\mathbb{O}} b), 0)$ and $\kappa_{\mathbb{S}}(R) = \|\mathcal{A}_{\mathrm{native}}(R)\|_2 / \|R\|_2^2 \in [0, 1]$ in 1 clock cycle.

2. 7-Bit Primitive Zero-Divisor Blade Register W_J[6:0]:
Selects active partner blade $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d)$ from 84 primitive blades ($c \in \{1..7\}$, $d \in \{9..15\}$, $d \neq c \oplus 8$, $s_d \in \{\pm 1\}$):
• C_IDX[2:0] $= c \in \{1..7\}$ (Lower Octonion Tap)
• D_IDX[3:0] $= d \in \{9..15\}$ (Upper Sedenion Tap)
• S_SIGN $= (s_d == -1)$? 1'b1 : 1'b0

3. Per-Slice Combinatorial Control Generation ($k \in \{0..15\}$):
• ADDR_A_k[3:0] $= k \oplus c$ (4-bit XOR Gate Array)
• ADDR_B_k[3:0] $= k \oplus d$ (4-bit XOR Gate Array)
• POL_A_k $=$ SIGN_NEG_KJ$[k][c]$ (16 × 16 256-Bit LUT ROM)
• POL_B_k $=$ S_SIGN $\oplus$ SIGN_NEG_KJ$[k][d]$

ELECTRICAL ENGINEERING & SIGNAL INTEGRITY SPECIFICATIONS (SHEET 1 OF 5):
1. Algebraic Zero-Multiplier Operation: Every channel slice SLICE_k ($k = 0..15$) implements the exact sedenion coordinate product $z_k(t) = \frac{1}{\sqrt{2}}\big(\sigma(k \oplus c, c)\, x_{k\oplus c}(t) + s_d\sigma(k \oplus d, d)\, x_{k\oplus d}(t)\big)$ purely via passive CMOS routing (ADG1606), differential wire-swap polarity inversion (ADG1611), and precision ±0.05% thin-film resistor summing (ERA-3AEB1001V + AD8130).
2. Switching Latency: Total blade reconfiguration time from U_CTRL GPIO update to settled analog output (0.01%) is t_{ON}(ADG1606) + t_{ON}(ADG1611) + t_{settle}(AD8130) = 140 ns + 115 ns + 45 ns = **300** ns (> 3.3 MHz zero-divisor blade hopping rate for ZDKM LPI/LPD waveforms).
3. Power Rails: Analog ±5.0 V ultra-low-noise rails (VCC_+5VA, VSS_-5VA) supplied by LT3045 / LT3094 LDOs (0.8 μV_{rms} noise); Digital +3.3 V logic rail (VDD_3V3) isolated via ferrite beads (BLM18PG121SN1D).

AGOURAKIS NATIVE SEDENION SIGNAL PROCESSING LAB.	CLASSIFICATION: OPEN HARDWARE / DARPA PS-26-09	
TOP-LEVEL 16-CH PRE-ADC ANALOG SEDENION CROSSBAR & ROM	DWG NO: SCH-NSSP-100-A	
DESIGNER: D. C. Agourakis | VERIFIED: Lean 4 (516 Thms) + SPICE MC (83.65 dB)	SIZE: ANSI B	SHEET 1 OF 5

STAGE 1: 16:1 XOR DIFFERENTIAL MULTIPLEXERS

STAGE 2: DPDT WIRE-SWAP POLARITY INVERTERS ($\sigma \in \{\pm 1\}$)

STAGE 3: ±0.05% THIN-FILM DIFFERENTIAL SUMMER & ANTI-ALIASING DRIVER

U1_k: 2× **ADG1606BRUZ** (16:1 Diff Mux A)

Selects Input Pair $i_A = k \oplus c$ ($R_{\text{on}} = 4.5\,\Omega$)

S1A..S16A (Pins 19..12, 4..11) ← V_IN[0..15]_P

DA (Pin 28)

S1B..S16B (Bank B Pins) ← V_IN[0..15]_N

A0..A3 (Pins 15,16,17,18) ← ADDR_A_k[3:0] $= k \oplus c$

EN (Pin 14) ← +3.3V (Always Enabled)

DB (Pin 28B)

VDD (Pin 13) $= +5.0$V, VSS (Pin 3) $= -5.0$V

GND (Pin 12) = AGND (100nF C0G C1_k, C2_k)

VA_MUX_P ($V^{+}_{k\oplus c}$)

VA_MUX_N ($V^{-}_{k\oplus c}$)

U3_k: **ADG1611BRUZ** (Quad SPST → DPDT Bridge A)

$R_{\text{on}} = 0.5\,\Omega,\ \Delta R_{\text{on}} = 0.08\,\Omega$ (16-Lead TSSOP)

SW1 (Pin 2-3): $\sigma = +1$

SW2 (Pin 7-6): $\sigma = -1$

SW3 (Pin 10-11): $\sigma = -1$

SW4 (Pin 15-14): $\sigma = +1$

IN1,IN4 (Pins 1,16) ← POL_A_k | IN2,IN3 (Pins 8,9) ← POL_A_k

POL_A_k = SIGN_NEG_KJ[k][c] $\in \{0,1\}$ via U5A_k (SN74LVC2G04)

R1_k: 1.00kΩ

±0.05% ERA-3A

R2_k: 1.00kΩ

±0.05% ERA-3A

R5_k: 1.41kΩ

SUM_P

SUM_N

U6_k: **AD8130ARZ** / **ADA4932-1**

270 MHz Differential Summing Amp (CMRR = 90 dB)

+IN (Pin 1)

-IN (Pin 8)

+OUT (Pin 6)

-OUT (Pin 3)

REF (Pin 4) ← VOCM (+1.0V ADC CM)

FB (Pin 5) ← R_FB = 707 Ω (ERA-3A)

+VS (Pin 7) = +5V, -VS (Pin 2) = −5V

R7_k: 49.9Ω

R8_k: 49.9Ω

C5_k: 33pF

V_ADC_k_P

V_ADC_k_N

R6_k: 1.41kΩ

U2_k: 2× **ADG1606BRUZ** (16:1 Diff Mux B)

Selects Input Pair $i_B = k \oplus d$ ($R_{\text{on}} = 4.5\,\Omega$)

S1A..S16A (Pins 19..12, 4..11) ← V_IN[0..15]_P

DA (Pin 28)

S1B..S16B (Bank B Pins) ← V_IN[0..15]_N

A0..A3 (Pins 15,16,17,18) ← ADDR_B_k[3:0] $= k \oplus d$

EN (Pin 14) ← +3.3V (Always Enabled)

DB (Pin 28B)

VDD (Pin 13) $= +5.0$V, VSS (Pin 3) $= -5.0$V

GND (Pin 12) = AGND (100nF C0G C3_k, C4_k)

VB_MUX_P ($V^{+}_{k\oplus d}$)

VB_MUX_N ($V^{-}_{k\oplus d}$)

U4_k: **ADG1611BRUZ** (Quad SPST → DPDT Bridge B)

$R_{\text{on}} = 0.5\,\Omega,\ \Delta R_{\text{on}} = 0.08\,\Omega$ (16-Lead TSSOP)

SW1 (Pin 2-3): +1

SW2 (Pin 7-6): −1

SW3 (Pin 10-11): −1

SW4 (Pin 15-14): +1

IN1,IN4 (Pins 1,16) ← POL_B_k | IN2,IN3 (Pins 8,9) ← POL_B_k

POL_B_k = S_SIGN ⊕ SIGN_NEG_KJ[k][d] via U5B_k

R3_k: 1.00kΩ

±0.05% ERA-3A

R4_k: 1.00kΩ

±0.05% ERA-3A

EXACT CIRCUIT NODE TRANSFER FUNCTION (SLICE k):

1. Tap A Polarity Output: $V_{\text{pol},A}(t) = \sigma(k \oplus c, c)\, V_{k\oplus c}(t)$, where $\sigma = (-1)^{\text{POL_A_k}} \in \{\pm 1\}$.
2. Tap B Polarity Output: $V_{\text{pol},B}(t) = s_d\, \sigma(k \oplus d, d)\, V_{k\oplus d}(t)$, where $s_d\sigma = (-1)^{\text{POL_B_k}} \in \{\pm 1\}$.
3. Summing Node Output ($R_1 = R_2 = R_3 = R_4 = 1.00\,\text{k}\Omega \gg R_{\text{on}}$):

$$V_{\text{ADC},k}(t) = \frac{1}{\sqrt{2}}\Big(\sigma(k \oplus c, c)\, V_{k\oplus c}(t) + s_d\, \sigma(k \oplus d, d)\, V_{k\oplus d}(t)\Big)$$

4. Zero-Divisor Jammer Nulling Condition: When $R(t) = U_J(t) = \frac{1}{\sqrt{2}}(e_a + s_b e_b)$ and $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d) \in \ker L_{\mathbb{S}}(U_J)$, the two active terms arrive with opposite signs and equal amplitudes: $V_{\text{pol},A}(t) + V_{\text{pol},B}(t) \equiv 0.0$ V prior to ADC quantization!

RESISTOR TOLERANCE & SWITCH IMPEDANCE BUFFERING ANALYSIS (SHEET 2 OF 5):

• Why $R_{\text{sum}} = 1.00\,\text{k}\Omega$ Eliminates CMOS Switch R_{on} Mismatch: The combined series on-resistance of **ADG1606** ($4.5\,\Omega$) + **ADG1611** ($0.5\,\Omega$) is $R_{\text{sw}} = 5.0\,\Omega$, which is absorbed into the nominal loop gain. Channel-to-channel on-resistance flatness is $\Delta R_{\text{on}} \leq 0.5\,\Omega$ (**ADG1606**) and $0.08\,\Omega$ (**ADG1611**). Across $R_{\text{sum}} = 1000\,\Omega$, a $0.5\,\Omega$ switch mismatch contributes only 0.05% gain error—matching the ±0.05% tolerance of the Panasonic **ERA-3AEB1001V** thin-film resistors (68.38 dB mean null depth; 82.76 dB with **LT5400-1** ±0.01% matched quad networks).

• Local Supply Decoupling per Slice: Each IC (**U1_k..U6_k**) includes local 100 nF 0402 C0G (**GCM155R71H104KE02D**) + 2.2 μF 0603 X7R bypass capacitors placed within 1.5 mm of **VDD**/**VSS** pins.

AGOURAKIS NATIVE SEDENION SIGNAL PROCESSING LAB.	CLASSIFICATION: OPEN HARDWARE / DARPA PS-26-09	
PIN-LEVEL SCHEMATIC: DIFFERENTIAL ANALOG CROSSBAR SLICE k	DWG NO: SCH-NSSP-101-A	
DESIGNER: D. C. Agourakis | VERIFIED: Lean 4 (516 Thms) + SPICE MC (83.65 dB)	SIZE: ANSI B	SHEET 2 OF 5

STAGE A: DYADIC 2×2 DUAL-POLARIZED TEM PATCH ARRAY ($\lambda/2 = 95.1$ mm @ L1)

ANT1: Dual-Linear Patch ($m = 0, n = 0$)

$\mathbf{r}_{0,0} = \left((0 - \frac{1}{2})\frac{\lambda}{2},\ (0 - \frac{1}{2})\frac{\lambda}{2},\ 0\right)$, XPD > 28 dB

`PORT_THETA` ($\mathbf{E}_\theta$) $\rightarrow$ `LNA1A (TQP3M9038) + SAW`

`PORT_PHI` ($\mathbf{E}_\phi$) $\rightarrow$ `LNA1B (TQP3M9038) + SAW`

ANT2: Dual-Linear Patch ($m = 0, n = 1$)

$\mathbf{r}_{0,1} = \left((0 - \frac{1}{2})\frac{\lambda}{2},\ (1 - \frac{1}{2})\frac{\lambda}{2},\ 0\right)$, XPD > 28 dB

`PORT_THETA` ($\mathbf{E}_\theta$) $\rightarrow$ `LNA2A (TQP3M9038) + SAW`

`PORT_PHI` ($\mathbf{E}_\phi$) $\rightarrow$ `LNA2B (TQP3M9038) + SAW`

ANT3: Dual-Linear Patch ($m = 1, n = 0$)

$\mathbf{r}_{1,0} = \left((1 - \frac{1}{2})\frac{\lambda}{2},\ (0 - \frac{1}{2})\frac{\lambda}{2},\ 0\right)$, XPD > 28 dB

`PORT_THETA` ($\mathbf{E}_\theta$) $\rightarrow$ `LNA3A (TQP3M9038) + SAW`

`PORT_PHI` ($\mathbf{E}_\phi$) $\rightarrow$ `LNA3B (TQP3M9038) + SAW`

ANT4: Dual-Linear Patch ($m = 1, n = 1$)

$\mathbf{r}_{1,1} = \left((1 - \frac{1}{2})\frac{\lambda}{2},\ (1 - \frac{1}{2})\frac{\lambda}{2},\ 0\right)$, XPD > 28 dB

`PORT_THETA` ($\mathbf{E}_\theta$) $\rightarrow$ `LNA4A (TQP3M9038) + SAW`

`PORT_PHI` ($\mathbf{E}_\phi$) $\rightarrow$ `LNA4B (TQP3M9038) + SAW`

Eθ_1 Eϕ_1 Eθ_2 Eϕ_2 Eθ_3 Eϕ_3 Eθ_4 Eϕ_4

STAGE B: PASSIVE 90°/180° QUADRATURE OCTONION HYBRID FEED
(Maps Circularly Polarized TEM Wavefields Directly onto Moreno Cone $V_2(\mathbb{R}^7)$)

LOWER OCTONION SUBSPACE $a = (e_1, \ldots, e_7) \in \mathrm{Im}(\mathbb{O})$ ($\mathbf{E}_\theta$ In-Phase Network)

• BAL1..BAL4 (`Mini-Circuits TC1-1-13M+` 180° Differential Baluns):
Splits the 4 $\mathbf{E}_\theta$ element ports into balanced differential pairs and Hadamard spatial mode combinations (Sum Σ, Elevation Δ_{el}, Azimuth Δ_{az}, Diagonal Δ_{diag}):

$e_1 = \frac{1}{2}(E_{\theta,1} + E_{\theta,2} + E_{\theta,3} + E_{\theta,4})$ (Monopulse Σ_θ)
$e_2 = \frac{1}{2}(E_{\theta,1} + E_{\theta,2} - E_{\theta,3} - E_{\theta,4})$ (Elevation $\Delta_{\mathrm{el},\theta}$)
$e_3 = \frac{1}{2}(E_{\theta,1} - E_{\theta,2} + E_{\theta,3} - E_{\theta,4})$ (Azimuth $\Delta_{\mathrm{az},\theta}$)
$e_4 = \frac{1}{2}(E_{\theta,1} - E_{\theta,2} - E_{\theta,3} + E_{\theta,4})$ (Quadrupole $\Delta_{q,\theta}$)
e_5, e_6, e_7: Quadrature spatial-gradient taps via `QCN-19+` ($\|a\|_2 = \|\mathbf{E}_\theta\|_2$).

UPPER SEDENION SUBSPACE $b = (e_9, \ldots, e_{15}) \in \mathrm{Im}(\mathbb{O})e_8$ ($\mathbf{E}_\phi$ +90° Quadrature)

• HYB1..HYB4 (`Mini-Circuits QCN-19+` 90° Quadrature Hybrids) + BAL5..BAL8:
Applies a passive $-90°$ quadrature phase shift to all 4 $\mathbf{E}_\phi$ ports prior to identical Hadamard balun combining:

$e_9 = \frac{1}{2}j(E_{\phi,1} + E_{\phi,2} + E_{\phi,3} + E_{\phi,4})$ (Monopulse Σ_ϕ)
$e_{10} = \frac{1}{2}j(E_{\phi,1} + E_{\phi,2} - E_{\phi,3} - E_{\phi,4})$ (Elevation $\Delta_{\mathrm{el},\phi}$)
$e_{11} = \frac{1}{2}j(E_{\phi,1} - E_{\phi,2} + E_{\phi,3} - E_{\phi,4})$ (Azimuth $\Delta_{\mathrm{az},\phi}$)
$e_{12}..e_{15}$: Matching quadrupole & spatial-gradient taps ($\|b\|_2 = \|\mathbf{E}_\phi\|_2$).

• Maxwell TEM Guarantee: For any circularly polarized incident wave ($|\mathbf{E}_\theta| = |\mathbf{E}_\phi|$), $\|a\|_2 = \|b\|_2$ and $\langle a, b\rangle = 0$ identically across all (θ, ϕ) ($\kappa_{\mathbb{S}} = 1.000000$)!

$a_1..a_7$ $b_1..b_7$

STAGE C: 16-CHANNEL PHASE-COHERENT IF DOWNCONVERTER
(Feeds `J1: V_IN0_P/N..V_IN15_P/N` on Sheet 1)

• Common LO Synthesizer (`U_PLL: Analog Devices ADF4351`):
Generates phase-locked $f_{\mathrm{LO}} = 1555.42$ MHz from 10 MHz OCXO reference, distributed via zero-skew 1:16 Wilkinson clock tree to all 16 active doubly-balanced mixers (`LTC5567`).

• Differential IF Output Stage ($f_{\mathrm{IF}} = 20.00$ MHz, BW = 10 MHz):
4th-order Butterworth differential bandpass filter + `ADA4940-2` differential IF driver outputting balanced 100 Ω differential signals `V_IN0_P/N..V_IN15_P/N` to Sheet 1 `J1`.

• Polarimetric Handedness Separation:

- LHCP Jammer / Multipath: Arrives at `J1` as $R_{\mathrm{jam}} = (a(\theta,\phi),\ +b(\theta,\phi)) \in V_2(\mathbb{R}^7) \implies$ Annihilated by W_J ($R_{\mathrm{jam}} \cdot W_J = 0$).
- RHCP Authentic Satellite: Arrives at `J1` as $R_{\mathrm{sat}} = (a(\theta,\phi),\ -b(\theta,\phi)) \in V_2(\mathbb{R}^7) \implies$ Amplified by +3.01 dB ($\|R_{\mathrm{sat}} \cdot W_J\|_2 = \sqrt{2}$).

STAGE D: ULTRA-LOW-NOISE ANALOG & DIGITAL POWER REGULATION

• `U_POS5V` (`Analog Devices LT3045EDD`, +5.0V 500 mA Ultra-Low-Noise LDO):
Input +6.5V $\rightarrow$ Output `VCC_+5VA` (+5.000V, 0.8 μV_{rms} noise 10 Hz–100 kHz, PSRR = 76 dB @ 1 MHz, $R_{\mathrm{SET}} = 50.0\,\mathrm{k\Omega}$ 0.1%, $C_{\mathrm{SET}} = 4.7\,\mu\mathrm{F}$).

• `U_NEG5V` (`Analog Devices LT3094EDD`, −5.0V 500 mA Ultra-Low-Noise Negative LDO):
Input −6.5V $\rightarrow$ Output `VSS_-5VA` (−5.000V, 0.8 μV_{rms} noise, PSRR = 74 dB @ 1 MHz, $R_{\mathrm{SET}} = 50.0\,\mathrm{k\Omega}$ 0.1%).

• `U_DIG3V3` (`ADM7150ACPZ-3.3`, +3.3V 800 mA Digital/GPIO Rail):
Powers `U_CTRL` (`XC7A35T`) and `U5_0..15` (`SN74LVC2G04`); star-grounded to `AGND` at `J2`.

RF FRONT-END & CALIBRATION-FREE MANIFOLD NOTES (SHEET 3 OF 5):
1. Zero Software Calibration Matrices: Because the passive 90° quadrature hybrids (`QCN-19+`) and 180° baluns (`TC1-1-13M+`) map the two orthogonal transverse field components ($\mathbf{E}_\theta, \mathbf{E}_\phi$) into $(a, b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})e_8$ with identical Hadamard spatial bases, the Moreno zero-divisor conditions $\|a\|_2 = \|b\|_2$ and $\langle a, b\rangle = 0$ hold by Maxwell's free-space TEM equations across all arrival angles (θ, ϕ).
2. Direct Baseband SDR Interface Option: For benchtop testing without the L1 RF patch array, `Stage A/B/C` can be bypassed by feeding 16 differential baseband DAC channels directly into `J1` on Sheet 1.

AGOURAKIS NATIVE SEDENION SIGNAL PROCESSING LAB.	CLASSIFICATION: OPEN HARDWARE / DARPA PS-26-09	
DYADIC 2x2 TEM VECTOR ARRAY & PASSIVE 90/180 OCTONION FEED	DWG NO: SCH-NSSP-102-A	
DESIGNER: D. C. Agourakis | VERIFIED: Lean 4 (516 Thms) + SPICE MC (83.65 dB)	SIZE: ANSI B	SHEET 3 OF 5

TABLE 1: COMPLETE BILL OF MATERIALS — 16-CHANNEL PRE-ADC ANALOG SEDENION CROSSBAR BOARD

Item	Ref Des	Manufacturer	Mfr Part Number	Pkg	Qty	Description / Electrical Specification
1	`U1_0..15, U2_0..15`	Analog Devices	`ADG1606BRUZ`	TSSOP-28	64	16:1 Low-R_{on} (4.5 Ω, ΔR_{on} = 0.5 Ω) CMOS Analog Multiplexer (2 per Diff Mux ×2 Taps ×16 Channels)
2	`U3_0..15, U4_0..15`	Analog Devices	`ADG1611BRUZ`	TSSOP-16	32	Quad SPST Ultra-Low-R_{on} (0.5 Ω, ΔR_{on} = 0.08 Ω) Configured as H-Bridge DPDT Polarity Reversal Switch
3	`U5_0..15`	Texas Instruments	`SN74LVC2G04DCKR`	SC70-6	16	Dual High-Speed CMOS Inverter (t_{pd} = 1.5 ns) for Complementary DPDT Switch Gate Drive (`POL` / `~POL`)
4	`U6_0..15`	Analog Devices	`AD8130ARZ`	SOIC-8	16	270 MHz Differential Receiver / Summing Amplifier (CMRR = 90 dB, Slew Rate = 1090 V/μs)
5	`R1_0..R4_15`	Panasonic	`ERA-3AEB1001V`	0603	64	1.00 kΩ ±0.05%, ±25 ppm/°C Precision Metal Thin-Film Summing Resistor (68.38 dB Mean Analog Null)
5A	`RN1_0..15 (Alt)`	Analog Devices	`LT5400AIMS8E-1`	MSOP-8	(32)	Optional ±0.01% Matched Quad 1.00 kΩ Precision Resistor Network (82.76 dB Mean Analog Null)
6	`R5_0..R6_15`	Panasonic	`ERA-3AEB1401V`	0603	32	1.40 kΩ ±0.05% Thin-Film Shunt Scaling Resistor (Sets Exact $1/\sqrt{2} \approx 0.7071$ Unit-Blade Normalization)
7	`R_FB_0..15`	Panasonic	`ERA-3AEB7060V`	0603	16	706 Ω ±0.05% Thin-Film Differential Feedback Gain Resistor
8	`R7_0..R8_15`	Yageo	`RT0603BRD0749R9L`	0603	32	49.9 Ω ±0.1% Anti-Aliasing Filter Series Output Resistor (100 Ω Differential ADC Termination)
9	`C5_0..15`	Murata	`GRM1885C1H330JA01D`	0603	16	33 pF ±5% 50V C0G/NP0 Differential Anti-Aliasing Filter Capacitor (f_{-3dB} = 48.3 MHz)
10	`C_BYP_1..192`	Murata	`GCM155R71H104KE02D`	0402	192	100 nF 50V X7R Local High-Frequency Supply Decoupling Capacitor (Placed ≤ 1.5 mm from IC Pins)
11	`U_ADC1, U_ADC2`	Texas Instruments	`ADS52J90ZZR`	BGA-144	2	8/16-Channel 14-Bit, 100 MSPS Simultaneous Phase-Coherent ADC (SNR = 71.5 dBFS, SFDR = 86 dBc)
12	`U_CTRL`	AMD / Xilinx	`XC7A35T-1CPG236C`	BGA-236	1	Artix-7 FPGA Implementing 256-Bit Cayley–Dickson Sign ROM, $K = 1$ Associator Detector & Q1.14 Beamformer
13	`U_POS5V`	Analog Devices	`LT3045EDD#PBF`	DFN-10	1	+5.0V, 500 mA, 0.8 μV_{rms} Ultra-Low-Noise Positive Analog LDO (PSRR = 76 dB @ 1 MHz)
14	`U_NEG5V`	Analog Devices	`LT3094EDD#PBF`	DFN-10	1	−5.0V, 500 mA, 0.8 μV_{rms} Ultra-Low-Noise Negative Analog LDO (PSRR = 74 dB @ 1 MHz)
15	`U_DIG3V3`	Analog Devices	`ADM7150ACPZ-3.3`	LFCSP-8	1	+3.3V, 800 mA Low-Noise Digital Logic LDO for `U_CTRL` and `U5_0..15`
16	`HYB1..HYB4`	Mini-Circuits	`QCN-19+`	1206	4	90° Passive Quadrature Hybrid (1.4–1.9 GHz, 0.3 dB Insertion Loss, ±1.5° Phase Balance)
17	`BAL1..BAL8`	Mini-Circuits	`TC1-1-13M+`	AT224	8	180° Passive Differential Flux-Coupled Balun (4.5–3000 MHz)
18	`J1, J2`	Samtec	`QSH-030-01-L-D-A`	QSH-60	2	60-Pin High-Speed Differential Mezzanine Connector (100 Ω Controlled Impedance)

TABLE 2: 256-BIT CAYLEY–DICKSON SIGN ROM CONTENTS (`SIGN_NEG_KJ[k][j]`)

Bit $j \in \{0, \ldots, 15\}$ (MSB-first, matching `logic [0:15][0:15] SIGN_NEG_KJ`) of Row k equals 1 iff $e_{k \oplus j} \cdot e_j = -e_k$:

```
Row k=00: 16'b0111111111111111 (16'h7FFF)   Row k=08: 16'b0000000001111111 (16'h007F)
Row k=01: 16'b0010100110010110 (16'h2996)   Row k=09: 16'b0101011000010110 (16'h5616)
Row k=02: 16'b0001110011000011 (16'h1CC3)   Row k=10: 16'b0110001101000011 (16'h6343)
Row k=03: 16'b0100101010100101 (16'h4AA5)   Row k=11: 16'b0011010100100101 (16'h3525)
Row k=04: 16'b0000011111110000 (16'h07F0)   Row k=12: 16'b0111100001110000 (16'h7870)
Row k=05: 16'b0101000110101010 (16'h51AA)   Row k=13: 16'b0010111000101010 (16'h2E2A)
Row k=06: 16'b0110010010011001 (16'h6499)   Row k=14: 16'b0001101100011001 (16'h1B19)
Row k=07: 16'b0011001011001100 (16'h32CC)   Row k=15: 16'b0100110101001100 (16'h4D4C)
```

Verified 256/256 bit-exact against Lean 4 (`Geometry.lean`) & RTL (`sedenion_pontryagin_cascade.sv`).

TABLE 3: 4-LAYER CONTROLLED-IMPEDANCE PCB STACKUP & SPICE MONTE CARLO VERIFICATION

PCB Layer Stackup (1.6 mm Total Thickness, Isola FR408HR / Rogers RO4350B Core):

- L1 (Top Signal, 1 oz Cu): 100 Ω edge-coupled differential analog pairs (W = 0.18 mm, S = 0.15 mm), `U1_k..U6_k`, `RN1_k`. Intra-pair length matching $\Delta L \leq 0.10$ mm (< 0.04° phase skew at 20 MHz IF).
- L2 (Solid Ground Plane, 1 oz Cu): Unbroken low-inductance `AGND` reference plane directly beneath all L1 differential traces (H_{12} = 0.20 mm prepreg).
- L3 (Split Power Plane, 1 oz Cu): `VCC_+5VA`, `VSS_-5VA` analog polygons under `SLICE_0..15`; isolated `VDD_3V3` digital polygon under `U_CTRL`.
- L4 (Bottom Digital Control, 1 oz Cu): 160-bit static/low-duty GPIO blade control lines `ADDR_A`, `ADDR_B`, `POL_A`, `POL_B` shielded from L1 analog summing nodes by L2 `AGND`.

SPICE Netlist Verification (`sedenion_analog_crossbar.cir`, 84 Blades × 200 Random Builds):
- ±1.0% Thick-Film (`RC0603`): 41.82 dB Mean Null | ±0.1% Thin-Film (`ERA-3AED`): 62.35 dB Mean Null
- **±0.05%** Thin-Film (`ERA-3AEB1001V`): 68.38 dB Mean Null (P_{10} = 59.23 dB, P_{90} = 78.30 dB)
- ±0.01% Bulk-Metal Foil (`LT5400-1`): 82.76 dB Mean Null (P_{10} = 74.29 dB, P_{90} = 93.13 dB)

FABRICATION, ASSEMBLY & TEST INSTRUCTIONS (SHEET 4 OF 5):
1. Companion Files in `hardware/schematic/`: Native KiCad 7/8 schematics (`sedenion_analog_crossbar.kicad_sch`, `sedenion_2stage_annihilating_cascade.kicad_sch`), SPICE3 netlists (`*.cir`), and 23-line CSV BoM (`BOM_PreADC_Analog_Crossbar.csv`; Stage-2 items 19–22 on Sheet 5).
2. Benchtop Bring-Up Self-Test: Set `W_J[6:0]` to Blade 0 ($c = 1, d = 10, s_d = +1$). Apply a 2.0 V_{pp} 10 MHz differential sine wave to `V_IN1_P/N` (e_1) and `V_IN10_P/N` (e_{10}) at `J1`. Verify on an oscilloscope at `J2` that all 16 differential outputs `V_ADC0..15_P/N` collapse to < 1.5 mV_{pp} (≥ 62 dB null), then invert `V_IN10_P/N` ($U_{sat} = (e_1 - e_{10})/\sqrt{2}$) and verify +3.01 dB coherent output restoration on channels $k = 3$ and $k = 8$.

AGOURAKIS NATIVE SEDENION SIGNAL PROCESSING LAB	CLASSIFICATION: OPEN HARDWARE / DARPA PS-26-09	
BILL OF MATERIALS (BOM), 256-BIT ROM & 4-LAYER PCB STACKUP	DWG NO: SCH-NSSP-103-A	
DESIGNER: D. C. Agourakis | VERIFIED: Lean 4 (516 Thms) + SPICE MC (83.65 dB)	SIZE: ANSI B	SHEET 4 OF 5

NON-ASSOCIATIVE PURE-ASSOCIATOR ANALOG FILTER: $y_{\text{out}}(t) = \big(R(t)\cdot w_1\big)\cdot w_2 = [R(t), w_1, w_2]$ WITH $w_1 \cdot w_2 = 0$ (LEAN 4 THEOREM 5.39)

STAGE 1: 2-TAP ANALOG CROSSBAR
(`SLICE_0..15`, $w_1 \in \mathcal{B}_{168}$)

- Input Field (100 Ω Diff):
$R(t) = \alpha(t)u_1 + \beta(t)u_2 + \gamma(t)s(t)$
Non-coaxial +80 dB jammers: $n(u_1) \perp n(u_2)$
(e.g., $u_1 = e_1 + e_{10}$, $u_2 = e_1 + e_{13}$, $s = e_2$).
- Stage-1 Blade Selection:
$w_1 = \frac{1}{\sqrt{2}}(e_4 - e_{15}) \in \ker L(u_1) \cap \mathcal{B}_{168}$
2× `ADG1606` + 2× `ADG1611` + `LT5400-1` (±0.01%).
- Stage-1 Output:
$R_1(t) = R(t)\cdot w_1 = \beta(t)(u_2 \cdot w_1) + \gamma(t)(s \cdot w_1)$
Jammer 1 annihilated pre-buffer (−83.65 dB mean null);
Jammer 2 rotated to $u_2' = \frac{1}{\sqrt{2}}(e_5 - e_2 - e_9 + e_{14})$.

32w

INTER-STAGE BUFFER BANK
(16× `ADL5565ACPZ`, 6 GHz)

- Impedance Isolation:
Prevents Stage-2 4-tap resistor array from loading Stage-1 `AD8130` feedback network.
- Key RF / IF Metrics:
– Bandwidth: **6.0** GHz (−3 dB)
– Diff Gain: A_V = +6.0 dB (strapped)
– Linearity: OIP3 = +**47** dBm @ 200 MHz
– Input Noise: **1.5** nV/$\sqrt{\text{Hz}}$
– Group Delay: < 0.25 ns (< 0.9° @ 10 MHz)
- Headroom Preservation:
Maintains > 78 dB SFDR entering Stage 2.

32w

STAGE 2: 4-TAP FOIL CROSSBAR
(`U_STG2_0..15`, $w_1 \cdot w_2 = 0$)

- Algebraic Collapse Obstruction Bypass:
In any associative algebra, $w_1 \cdot w_2 = 0 \implies (x \cdot w_1)\cdot w_2 \equiv 0$ $(\forall x)$.
In $\mathbb{S}$, non-associativity turns $w_1 \cdot w_2 = 0$ into the pure associator:
$y_{\text{out}}(t) = (R(t)\cdot w_1)\cdot w_2 - R(t)\cdot(w_1 \cdot w_2) = [R(t), w_1, w_2]$.
- 4-Tap Compound Blade Hardware (`LT5400AIMS8E-1`):
$w_2 = \frac{1}{2}(e_1 - e_6 + e_{10} - e_{13}) \in \ker R(u_2 \cdot w_1) \cap \ker L(w_1)$.
– 4× `ADG1606` + 4× `ADG1611` per channel slice k
– 1× `LT5400AIMS8E-1` quad matched 4 × 1.00 kΩ (±0.01%, 0.2 ppm/°C) per rail + `AD8130` (R_{FB} = 500 Ω exact $\frac{1}{2}$ gain).
- Simultaneous Dual-Jammer Null + Signal Gain:
$((u_1 \text{ or } u_2)\cdot w_1)\cdot w_2 = 0$ (−78.51 dB SPICE MC); $\|[e_2, w_1, w_2]\|_2 = 1/\sqrt{2}$ (unit blades).

ADC

RFSOC PONTRYAGIN CONTROLLER
(`XCZU48DR`, < 200 ns LOOP)

- RTL Core (`sedenion_pontryagin_cascade.sv`):
– 2-cycle `Q1.14` switching vector $\psi = \bar{x}\cdot p$ (3.3 ns @ 300 MHz)
– Exact 3-term $\ker L(u)$ projector ($\|u \cdot w^\star\|_2 = 0.00$)
– 168-blade bang-bang argmax $w^\star = \text{argmax}_{b\in\mathcal{B}_{168}}\langle b, \psi\rangle$
– Hardware flag `w1_w2_mutually_annihilating`
– Extremal $[16, 8, 6]_3$ self-dual ternary FEC ($d_{\min} = 6$).
- End-to-End Closed-Loop Latency (< 195 ns):
RFSoC Logic (< 15 ns) + GPIO (4 ns) + `ADG1606/1611` t_{ON} (118 ns) + `ADL5565/AD8130` Settling (< 25 ns) = **162** ns < 200 ns.

480-BIT CLOSED-LOOP GPIO BLADE BUS: `CTRL_STG1[159:0]` (w_1) + `CTRL_STG2[319:0]` (w_2) — RECONFIGURES IN < 195 ns

TABLE 4: STAGE-2 MUTUALLY-ANNIHILATING CASCADE & RFSOC ADD-ON BOM (ITEMS 19–22)

Item	Ref Des	Mfr	Part Number	Pkg	Qty	Function / Spec
19	`U_BUF_0..15`	Analog Devices	`ADL5565ACPZ-R7`	LFCSP-16	16	6 GHz Diff Inter-Stage Buffer (OIP3 = +47 dBm, 1.5 nV/$\sqrt{\text{Hz}}$)
20	`RN_STG2_0..31`	Analog Devices	`LT5400AIMS8E-1`	MSOP-8	32	Quad 1.00 kΩ ±0.01% (0.2 ppm/°C) 4-Tap w_2 Foil Network
20A	`R_FB2_0..15`	Panasonic	`ERA-3AEB501V`	0603	16	500 Ω ±0.05% Stage-2 Feedback Resistor (Exact $\frac{1}{2}$ Blade Gain)
21	`U_ADCDRV_0..15`	Analog Devices	`ADA4945-1ACPZ`	LFCSP-16	16	145 MHz Low-Distortion (−115 dBc HD2/HD3) Diff ADC Driver
22	`U_RFSOC`	AMD / Xilinx	`XCZU48DR-2FSVG1517E`	FSVG1517	1	Zynq UltraScale+ Gen-3 RFSoC (< 200 ns Pontryagin + Cascade RTL)

Complete Companion Design Files (`hardware/schematic/` & `benchmarks/gps_jamming/rtl/`):
- `sedenion_2stage_annihilating_cascade.cir` (SPICE3) | `sedenion_2stage_annihilating_cascade.kicad_sch` (KiCad 7/8)
- `sedenion_pontryagin_cascade.sv` (SystemVerilog) | `crossbar_84_blades_rom.json` (14 Box-Kite Sextets + J_3 ROM)

TABLE 5: SPICE MONTE CARLO ($N = 200$ BUILDS) & LIVE SILICON VERIFICATION

1-Stage vs. 2-Stage Dual-Jammer Analog Null Depth (`generate_crossbar_netlist.py`):

- ±0.05% Thin-Film (`ERA-3AEB1001V`): 68.23 dB (1-Stage) | 64.54 dB (2-Stage Dual Jammer)
- **±0.01%** Matched Foil (`LT5400AIMS8E-1`): 83.65 dB (1-Stage) | 78.51 dB (P_{10} = 72.18 dB, P_{90} = 85.08 dB)
- ±0.001% Laser-Trimmed Monolithic IC: 102.12 dB (1-Stage) | 98.06 dB (2-Stage Dual Jammer)

Live 240 MHz ESP32-S2 Xtensa LX7 Silicon (`/dev/cu.usbmodem01`, `sedenion selftest`):
- [1/5] N_u Calculus: 1712 μs, $\|L(u)^{-1}L(u) - I\|_\infty = 3.74 \times 10^{-7}$, Moore–Penrose err = 1.12×10^{-7}
- [2/5] 2-Stage Cascade ($w_1 \cdot w_2 = 0$): 1287 μs, $\|w_1 \cdot w_2\| = 0.0$, Dual-Jammer Out = 0.0, $\|[e_2, w_1, w_2]\| = 2.0000$
- [3/5] Pontryagin PMP + $\mathcal{B}_{168}$: 2176 μs, $H^\star = 5.4201$, `B168:id=69(-1)` (3336 μs), leak = 0.00
- [4/5] S_3 & Sextic $J_3(u)$: $J_3(e_1) = +1.00$, $J_3(e_9) = -1.00$, $|J_3(u) - J_3(\psi(u))| = 1.79 \times 10^{-7}$
- [5/5] Hurwitz–Radon 9-Frame & $[16, 8, 6]_3$ FEC: ZD err = 1.49×10^{-8}, 513/513 cosets PASS (`ALL_5_PILLARS_PASS`)

2-STAGE ANNIHILATING CASCADE BRING-UP NOTES (SHEET 5 OF 5):
1. Mode A (Antipodal Jammers, $u_2 = \bar{u}_1$): Both w_1 and w_2 are 2-tap primitive blades (2,688 hits in $\mathcal{B}_{84} \times \mathcal{B}_{84}$); unused taps C, D in Stage 2 are disabled via `ADG1606 EN=0`.
2. Mode B (Orthogonal-Axis Non-Coaxial Jammers, $n(u_1) \perp n(u_2)$, Theorem 5.39): All 4 matched resistors of `LT5400AIMS8E-1` sum $w_2 = \frac{1}{2}(e_1 - e_6 + e_{10} - e_{13})$, achieving 78.51 dB pre-ADC suppression across both independent +80 dB jammers simultaneously with zero adaptive matrix inversion.

AGOURAKIS NATIVE SEDENION SIGNAL PROCESSING LAB. CLASSIFICATION: OPEN HARDWARE / DARPA PS-26-09		
2-STAGE MUTUALLY-ANNIHILATING CASCADE & RFSOC PONTRYAGIN LOOP	DWG NO: SCH-NSSP-200-A	
DESIGNER: D. C. Agourakis | VERIFIED: Lean 4 (516 Thms) + SPICE MC (83.65 dB)	SIZE: ANSI B	SHEET 5 OF 5